A-A-53639
6 September 1988

COMMERCIAL ITEM DESCRIPTION

WAX, DENTAL (Sticky)

The General Services Administration has authorized the use of this Commercial Item Description in preference to Interim Federal Specification U-W-00149A (DSA-DM).

Salient characteristics:

This commercial item description covers sticky wax used for holding various molding materials in dental prosthetic procedures.

The sticky wax shall be a mixture of waxes and resins or other ingredients combined in proportions that will result in a material which may be used in prosthetic dentistry. The wax shall be furnished in smooth sticks 2 7/8 to 4 1/8 inches in length. The cross section shall be round or hexagonal and shall be 1/4 ± 1/16 inch in diameter, or 1/4 ± 1/16 inch across the flats. The color shall be dark red, orange, or dark violet so as to contrast with the white, tan, gray, or pink surfaces of artificial dental stones and plaster to which the wax is applied.

Performance. The wax shall be sticky and have good adhesive characteristics when melted. The wax shall be brittle at room temperature and fracture cleanly when broken.

Flow. The flow shall be not more than 5.0 percent at 30 ± 0.1 °C and not less than 60 percent at 43 ± 0.1 °C when tested as specified under flow in test section.

Ash. The ash content shall be no greater than 0.2 percent when tested as specified under ash in test section.

Shrinking. The wax shall shrink no more than 0.65 percent when cooled from 43°C to 28°C when tested as specified under shrinking in test section.

Lot or batch number. Each unit package shall be marked with a lot or batch number and date of manufacture.

Delivery Schedule. Not more than three months shall have elapsed from date of manufacture of the wax to date of delivery to the Government.

AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:22Z
Check the source to verify that this is the current version before use.

Workmanship. The wax shall be furnished clean and free from defects which detract from its appearance or impairs its serviceability.

Unit. Package (PG). Two ounces of wax, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one 2 oz. box of sticky wax. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests shall be conducted at inspection level S-1. For tests, acceptance number shall be zero (0) for all sample sizes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:22Z
Check the source to verify that this is the current version before use.

Examination. Examination shall include but shall not be limited to the following.

Color and weight not as specified.
Wax not free of foreign material.

Tests. Tests shall be conducted to determine compliance with specification requirements.

Preparation of test specimens. A quantity of wax shall be broken into pieces and placed in a metal pouring pan. The pan shall be placed on a surface which is 130 millimeters (mm) (5.1 inches) below a 250-watt infrared lamp. The wax, while being stirred, shall be allowed to reach a temperature of 75 ± 5°C (167 ± 9°F) and be maintained at this temperature until the sample is melted throughout. A thermometer shall be used to measure the temperature. The melted wax shall then be poured into a mold that has been lubricated with a silicone grease whose melting point is higher than 75 ± 5°C (167 ± 9°F). The mold shall consist of a stainless steel plate 6.0 mm (0.236 inches) thick, having flats, parallel top and bottom surfaces, and containing four holes 10.0 mm (0.394 inches) in diameter. The axes of the holes shall be perpendicular to the surfaces of the plate. The sides of the holes shall be finished smooth. The mold shall be preheated to a temperature of 55 ± 5°C (131 ± 9°F) and placed on a smooth glass slab 152 mm long, 76 mm wide and 19 mm thick (6 X 3 X 0.75 inches) preheated to the same temperature. As the wax freezes and a shrinkage void appears, liquid wax shall be added. When the wax has lost its "mirror-like" surface, a smooth, flat tin-foil or aluminum-foil covered glass plate, preheated to 55 ± 5°C (131 ± 9°F), shall be placed on the top of the mold. A load of 9,000 grams (20 pounds) shall be applied to the top of the foil-covered glass plate for 30 minutes. The weight and the glass plate shall then be removed and the excess wax trimmed away. This may be accomplished by drawing the specimen flush with the surface. The mold shall be removed from the glass slab by gently-tapping the side of the mold. The specimens of wax shall be removed from the mold by chilling in water at 10°C, (50°F) and shall then be stored at 23.0 ± 2.0°C. (73.4 ± 3.6°F) for 24 hours before testing.

Flow. The initial length of the specimen shall be determined at 23.0 ± 2.0°C (73.4 ± 3.6°F) using a metric micrometer caliper. Four measurements shall be made around the circumference and one measurement shall be made in the center of the specimen. The measurements shall be averaged and recorded to the nearest 0.005 mm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:22Z
Check the source to verify that this is the current version before use.

The specimen and flow testing instrument shall be placed in a water bath and held at the testing temperature for 20 minutes prior to testing. The flow testing instrument shall consist of the following: A, a weight; B, a shaft having low thermal conductivity; and C, a brass platen. The total weight, in air, of these three components shall be 2,000 grams (4.41 pounds). The weight, A, shall be separated a minimum distance of 76 mm (3 inches) from the brass platen by the non-conducting shaft. This shaft shall be of hard rubber or a similarly poor thermal conductor to reduce loss of heat from the specimen. The diameter of the brass platen shall not be less than 51 mm (2 inches). The thickness shall not exceed 6.35 mm (0.25 inches). The temperature of the bath shall be controlled to within $\pm$ 0.1°C ($\pm$ 0.2°F) of the required temperature, a calibrated thermometer being used for determining the temperature. Agitation shall be provided in the water bath by means of a mechanical stirrer. A thin sheet of waterproof cellophane shall be placed between the instrument and each end of the specimen. The bottom of the specimen shall be 51 mm (2 inches) below the surface of the water in the bath. A constant axial load of 2,000 grams (4.41 pounds) shall then be applied to the specimen for 10 minutes.

Ash. A one-gram specimen of wax shall be ignited in a porcelain crucible at 930 to 950°F. and the residue calculated. The crucible shall be previously ignited to constant weight.

Shrinking. A specimen 20 centimeters in length shall be conditioned for 20 minutes in water at a temperature of 43°C. The water shall be cooled gradually to 28°C, and the change in length determined by observation with a comparator microscope.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:22Z
Check the source to verify that this is the current version before use.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

National Stock Number	Item Identification
6520-00-567-4000	WAX, DENTAL (Sticky)

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

48213

Project No. 6520-2094

Location: NCR Disk #CID6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:22Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS. In a continuing effort to make our standardization documents better, the DoD provides this form for use in ...itting comments and suggestions for improvements. All users of military standardization documents are invited to provide s... strons. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 4986 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
ATTN. DPSC-ASTE
2800 South 20th Street
Philadelphia, Pennsylvania 19101-8419

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:22Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
A-A-53639	WAX, DENTAL (Sticky)

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City State ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last First MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:22Z
Check the source to verify that this is the current version before use.